

Shoumik Saha

 [shoumiksaha.github.io](https://github.com/shoumiksaha)

 smksaha@umd.edu

 [in/shoumik-saha](https://in.linkedin.com/in/shoumik-saha)

 [/ShoumikSaha](https://github.com/ShoumikSaha)

 [Google Scholar](https://scholar.google.com/citations?user=smksaha)

RESEARCH INTEREST

- **Reliable AI** - Security & Reliability of GenAI, LLM Alignment & Hallucination
- **Computer Security** - Adversarial Attacks & Defenses, Malware

WORK EXPERIENCE

Applied Scientist Intern

Amazon AWS

Managers - [Varun Kumar](#), [Zijian Wang](#)

May 2025 – Aug 2025

Full Time

Graduate Research Assistant

Department of Computer Science

[University of Maryland](#)

Supervisor - [Dr. Soheil Feizi](#)

Aug 2023 – Present

Full Time

Applied Scientist Intern

Amazon AWS

Manager - [Zijian Wang](#)

Jun 2024 – Aug 2024

Full Time

Graduate Research Assistant

Maryland Cybersecurity Center

Supervisor - [Dr. Tudor Dumitras](#)

Aug 2022 – Jul 2023

Full Time

Lecturer

Department of Computer Science and Engineering

United International University

Jul 2021 – Jul 2022

Full Time

Research Assistant

Data Science and Engineering Research Lab

Bangladesh University of Engineering and Technology (BUET)

Mar 2021 – Jul 2022

Part Time

EDUCATION

University of Maryland - College Park (UMD)

Ph.D. in Computer Science (Ongoing)

Advisor – [Dr. Soheil Feizi](#)

Aug 2022 – Present

CGPA – 3.814/4.0

University of Maryland - College Park (UMD)

M.S. in Computer Science

Advisor – [Dr. Soheil Feizi](#)

Aug 2022 – Aug 2024

Bangladesh University of Engineering and Technology (BUET)

B.Sc. in Computer Science and Engineering

Thesis supervisor – [Dr. Atif Hasan Rahman](#)

Mar 2016 – Feb 2021

Advanced GPA – 3.79/4.0

PUBLICATIONS

1. **Under the Hood of SKILL.md: Semantic Supply-chain Attacks on AI Agent Skill Registry**
[Shoumik Saha, K Faghih, S Feizi]
[Under Review](#)([arXiv link](#))
Media Coverage: [The Register](#)
2. **Breaking the Code: Security Assessment of AI Code Agents Through Systematic Jailbreaking Attacks**
[Shoumik Saha, J Chen*, S Mayers*, SK Gouda, Z Wang, V Kumar]
[Under Review](#)([arXiv link](#))

3. **Almost AI, Almost Human: The Challenge of Detecting AI-Polished Writing**
[Shoumik Saha, S Feizi]
[ACL 2025](#) (Association for Computational Linguistics) ([Paper Link](#))
Media Coverage: [The New York Times](#), [Plagiarism Today](#), [The Science Archive](#), [The Cheat Sheet](#)
4. **Adversarial Paraphrasing: A Universal Attack for Humanizing AI-Generated Text**
[Y Cheng, VS Sadasivan, Shoumik Saha*, M Saberi*, S Feizi]
[NeurIPS 2025](#) (Conference on Neural Information Processing Systems) ([Paper Link](#))
5. **ProcVQA: Benchmarking the Effects of Structural Properties in Mined Process Visualizations on Vision–Language Model Performance**
[KT Zinat*, SM Abrar*, Shoumik Saha, S Duppala, SN Sakhamuri, Z Liu]
[EMNLP 2025](#) (Empirical Methods in Natural Language Processing) ([Paper Link](#))
6. **LLM-Check: Investigating Detection of Hallucinations in Large Language Models**
[G Sriramanan, S Bharti, VS Sadasivan, Shoumik Saha, P Kattakinda, S Feizi]
[NeurIPS 2024](#) (Conference on Neural Information Processing Systems) ([Paper Link](#))
7. **Fast Adversarial Attacks on Language Models In One GPU Minute**
[VS Sadasivan, Shoumik Saha*, G Sriramanan*, P Kattakinda, A Chegini, S Feizi]
[ICML 2024](#) (International Conference on Machine Learning) ([Paper Link](#))
Media Coverage: [The Register](#)
8. **DRSM: De-Randomized Smoothing on Malware Classifier Providing Certified Robustness**
[Shoumik Saha, W Wang, Y Kaya, S Feizi, T Dumitras]
[ICLR 2024](#) (International Conference on Learning Representations) ([Paper Link](#))
9. **MAlign: Explainable Static Raw-byte Based Malware Family Classification using Sequence Alignment**
[Shoumik Saha, S Afroz, A Rahman]
[Computers & Security Journal](#) ([Paper Link](#))
10. **ML-Based Behavioral Malware Detection Is Far From a Solved Problem**
[Y Kaya, Y Chen, Marcus Botacin, Shoumik Saha, F Pierazzi, L Cavallaro, D Wagner, T Dumitras]
[SaTML 2025](#) (Secure & Trustworthy Machine Learning) ([Paper Link](#))

SCHOLARSHIPS & AWARDS

- Awarded 'Dean's Fellowship' from the University of Maryland
- Awarded 'Innovation Fund' for research from the ICT division of Bangladesh government
- Achieved 'Merit Stipends' from BUET in five out of seven terms
- Awarded 'Dean's Award' in Junior year for extra-ordinary result
- Achieved 'Talent-Pool Scholarship' in High School

TEACHING

Teaching Assistant

University of Maryland

Fall 2023

Data 200: Knowledge in Society: Science, Data and Ethics

Lecturer

United International University

Fall 2021 - Summer 2022

Discrete Mathematics, Data Structure and Algorithm, Operating Systems

SERVICES

- **Reviewer**, top-tier ML conferences such as ICML 2024, ICLR 2025, NEURIPS 2025, etc.
- **External Reviewer**, top-tier Computer Security conference, including USENIX 2023
- **Lab Instructor**, United International University
- **General Secretary**, BUET Photographic Society